

ADS6311 DATASHEET

SPAD Depth Sensor for LiDAR Application

Revision history

Date	Revision	Changes
2023.03.04	V0.1	Initial release
2023.03.29	V0.2	1. Fix the ADR0/ADR1/ADR2 typo 2. Update 7.2 Recommended Operating Conditions (Power and Ground, and power combination suggestion) 3. Update 7.3 Current characteristics (Typical and Max Current) 4. Update 7.5 Digital input and output (IOV18 and IOV33 signals) 5. Update 7.2 Recommended Operating Conditions (Separate SV11 and AV11)
2023.04.20	V0.30	1. Add "System connection example" chapter.
2023.05.31	V0.31	1. Modify the digiPOT to Power IC in "5 Block Diagram" chapter. 2. Add default value notes in "10.1.5 I2C device address" chapter.
2023.06.05	V0.32	1. Modify the ADC_VIN[0] 和 ADC_VIN[1] connection in SoC and FPGA scheme block diagram in "System connection example" chapter.
2023.07.20	V0.33	1. Correct the "2D scanning mode timing" diagram.
2023.08.24	V0.34	1. Correct the SSYNC typo. 2. Add I_{AV11} in "Current characteristics".
2023.09.04	V0.35	1. Correct the DRVE/TRIG_Ox/TRIG_Fx signals in "FPGA scheme block diagram" diagram.
2023.09.05	V0.36	1. Modify the power supply value in "Recommended Operating Conditions" chapter.
2023.09.28	V0.4	1. Fix some typos. 2. Update the "Current characteristics" chapter.
2023.10.19	V0.5	1. Add "Electrostatic rank" features 2. Update the "DC Characteristics" chapter 3. Update the "Current Characteristics" chapter 4. Update the "AC Characteristics" chapter. 5. Add "SPAD Characteristics" chapter.
2023.11.22	V0.51	1. Add Robot, ADAS and other application filed in introduction chapter. 2. Correct the pin name TRGI to TRIG_I.
2023.12.06	V0.52	1. Modify the "Figure 1-1 the ROI for declining light strip" for Avoiding ambiguity.
2023.12.22	V0.53	1. Correct IOV33 to IOV18 in "Figure 1-2 TRIG_I Waveform schematic diagram". 2. Add VOP Typical and Max voltage in "Recommended Operating Condition" chapter. 3. Correct 22Mcps to 35Mcps and 45Mcps to 76Mcps in "Operating average current" chapter. 4. Modify APP=1% @Typ in "SPAD Characteristic" chapter.
2024.01.26	V0.54	1. Modify the SPAD PDE and VOP values in "SPAD Characteristics" chapter. 2. Add VC0 and VC1 transmission sequence description in "MIPI interface" chapter.

2024.03.11	V0.6	1. Update the chip size and optical center diagram. 2. Add BGA pin list top view in "PIN description" chapter. 3. Fix the typo from HROLL4 to HROLL3 in "System connection example".
2024.06.20	V0.61	1. Modify the SPAD After Pulse to 1%@typ. and 5%@Max in "SPAD Characteristics" chapter. 2. Add Operating Temperature and Storage temperature information in "Features" chapter. 3. Add Typ. ⁽⁰⁾ test condition in "Operating average current" chapter.
2024.06.28	V0.62	1. Add "Ordering Information" Chapter. 2. Modify "Wafer thickness after grinding" from 250um to 350um.
2024.08.14	V0.63	1. Fix the typo error in "power up sequence" chapter. 2. Revise the SPAD trigger rate and give the CPS calculation formula in "Operating average current" chapter.
2024.10.25	V0.64	1. Adding physical pixel (0, 0) and pixel (255, 191) mark in the "chip size and optical center" diagram.
2024.12.20	V0.65	1. Update BGA Pin list information and Figure in chapter 4.2 2. Add BGA package reflow information in chapter 7.1. 3. Update VOP typical value in chapter 7.2. 4. Update SPAD VBD and VBD vs T characteristic in chapter 7.6. 5. Add Temperature sensor characteristics in chapter 7.7. 6. Add ADC characteristics in chapter 7.8. 7. Update ordering information for new IBG Package in chapter 12. 8. Update datasheet contents. 9. Update some text formats.
2025.03.06	V0.66	1. Update electrostatic rank status in Features. 2. Update Test SPAD Pin in Bare Die list and IOV33 in BGA MAP. 3. Update VOP range in chapter 7.1 & chapter 7.2. 4. Update Current Characteristics in chapter 7.4. 5. Update SPAD Characteristics in chapter 7.6.
2025.05.30	V0.67	1. Update SPAD Characteristics in chapter 7.6 2. Modify the max value of ADC_VIN to ADC_Vref in chapter 7.8.
2025.07.17	V0.68	1. Update Standby current in chapter 7.4.1. 2. Update SPAD Characteristics in chapter 7.6. 3. Update ordering information in chapter 12.
2025.08.10	V0.69	1. Update SPAD Characteristics in chapter 7.6. 2. Update VOP voltage chapter 7.4.2.

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1 Introduction

The ADS6311 is a dTOF SPAD Image Sensor up to 256(H) x 192(V) pixels resolution. This chip has four imaging modes (gray scale mode, histogram mode, echo mode, ranging mode) and supports 1D and 2D scanning modes. The ADAPS unique leading on-chip DSP architecture and algorithm is integrated which improve the ranging precision and power consumption. Up to 5 echoes are supported to meet the performance and complex scenarios required by LiDAR.

Applications: LiDAR for automotive, Robot, ADAS, Security Monitoring, Dimensioning, Fall Prevention, etc.

2 Features

■ Image Array

- 768(H) x 576(V) SPAD Array, more than 442K SPAD
- Support up to 256(H)x192(V) pixel resolution @MP=3*3 SPAD binning in Ranging Mode
- Support full SPAD Array 768(H) x 576(V) resolution in Gray Scale Mode

■ SPAD features

- Pitch 10.08*10.08 um
- Passive quench, SPAD deadtime: 7ns~20ns
- SPAD PDE: 26% @940nm

■ Scanning mode

- Rolling shutter, up to 256(H) x 6(V) pixels per rolling
- Support 1D (vertical direction only) and 2D (vertical and horizontal direction) scanning mode
- Support flexible ROI (Region of Interest) configuration and various light strip shape

■ TDC features

- Typical TDC resolution 1ns
- TDC bin number=672

■ Laser control features

- Support anti-interference
- Support laser pulse width and period configuration
- Up to 4 VCSEL emission trigger signals (TRIG_O0/1/2/3) and feedback signals (TRIG_F0/1/2/3)

- Support external emission trigger signal (TRIG_I)

■ Communication Interface

- I2C: 1MHz, 400KHz
- SPI: 30MHz (I2C and SPI cannot support concurrently, use CSEL pin to select which one)
- MIPI: 4 Lanes, up to 1.5Gbps/lane; 2 VCs (VC0 and VC1)
- Support multiple sensors synchronization with FSYNC and SSYNC

■ Output mode

- Gray Scale mode (Intensity mode), up to 768(H) x 576(V) resolution
- Ranging mode, depth value and noise information are output
- Echo mode, up to 5 echoes raw histogram are output
- Histogram mode, raw histogram data are output

■ Safety function feature

- Support MBIST
- Support logic and data path self-detection at chip initial stage
- Support Voltage and temperature monitor
- Support clock monitor
- Support VCSEL trigger monitor for eyes protection

■ Temperature

- Operating temperature: Ta= -40°C to 105°C, Tj= -40°C to 125°C
- Storage temperature: Ta= -40°C to 125°C
- AEC-Q100 grade 2(TBD)

■ Package

- Package: COB/BGA

■ Electrostatic rank

- HBM: $\geq \pm 2\text{kV}$
- CDM: $\geq \pm 500\text{V}$
- Latch-Up: $\pm 200\text{mA}$

3 Chip Size and Optical Center

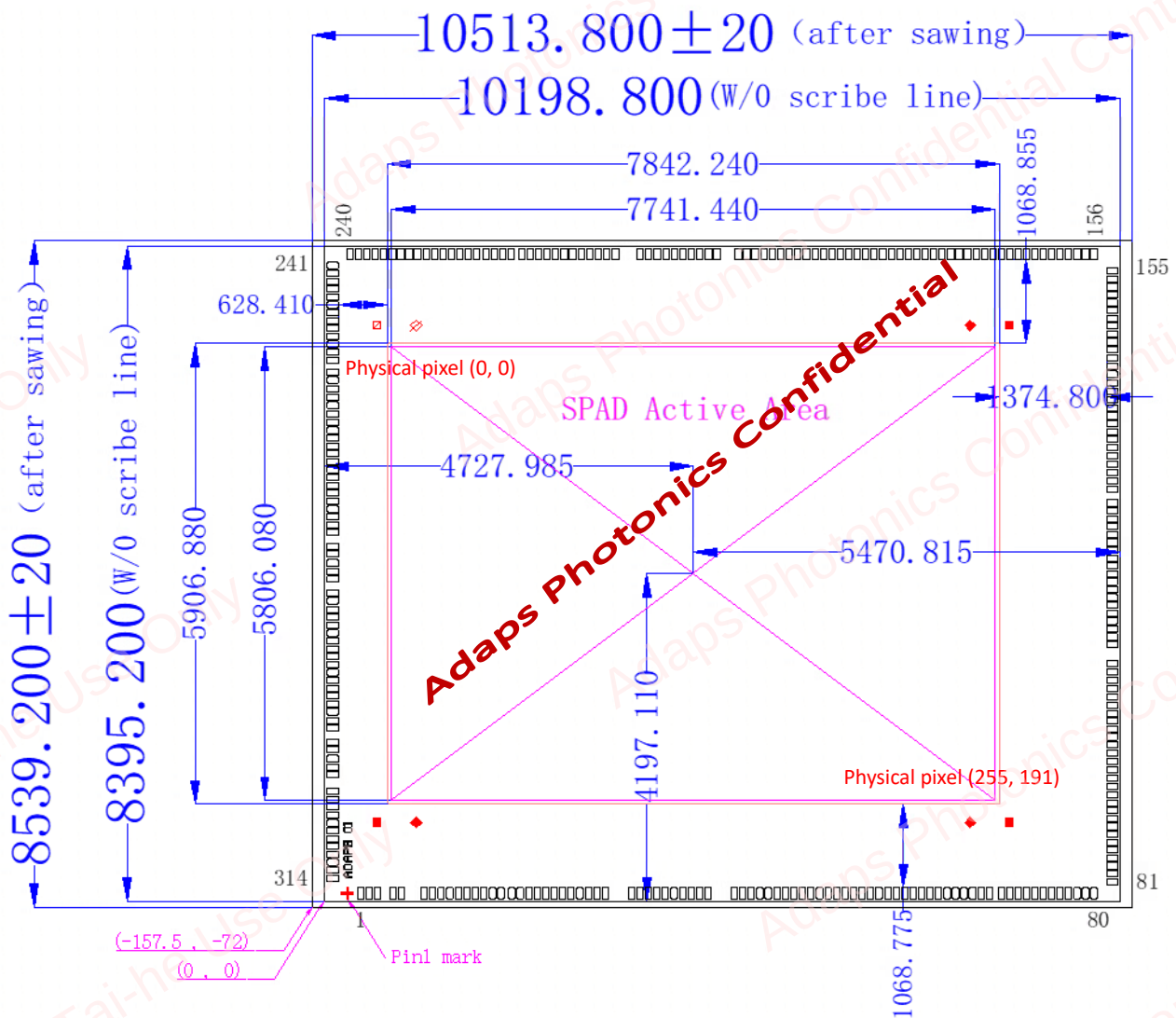


Figure 3-1 Chip Size and Optical Center (Top View)

Note:

- Unit: μm
- Die size (without scribe line): $10198.8 \times 8395.2 \mu\text{m}$
- Actual Die Size (After sawing to dice): $10513.8 \pm 20 \times 8539.2 \pm 20 \mu\text{m}$
- SPAD Active area size: $7741.44 \times 5806.08 \mu\text{m}$
- SPAD Visual area size: $7842.24 \times 5906.88 \mu\text{m}$
- SPAD area center: $(4727.985, 4197.110)$
- Physical pixel (0, 0): adjacent to pin 241 (chip left upper corner)

- Pin Size = 80um*140um, Pin Pitch >= 100um
- Scribe line width: X=355um, Y=184um
- PAD number: 314
- PAD opening: 80 x 140 um
- Min. pad pitch: 108um
- Wafer thickness after grinding: 350um (w/o uLens), uLens has ~5um height

4 PIN description

4.1 Bare Die/COB pin list

Table 4-1 Pin List

Pin num	Pin Name	Location	I/O	Pin Function
1	DV11	bottom	power	Digital 1.1V Power Supply
2	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
3	IOV18	bottom	power	Digital IO 1.8V Power Supply
4	DV11	bottom	power	Digital 1.1V Power Supply
5	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
6	DV11	bottom	power	Digital 1.1V Power Supply
7	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
8	IOV18	bottom	power	Digital IO 1.8V Power Supply
9	SV11	bottom	power	SPAD Array 1.1V Power Supply
10	SVSS11	bottom	ground	SPAD Array 1.1V GND
11	VOP	bottom	power	SPAD Anode negative high voltage supply Vop
12	SV33	bottom	power	SPAD Array Vex power supply
13	SVSS33	bottom	ground	SPAD Array Vex GND
14	DV11	bottom	power	Digital 1.1V Power Supply
15	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
16	NC	bottom	I/O	Not Connected
17	NC	bottom	I/O	Not Connected
18	DV11	bottom	power	Digital 1.1V Power Supply
19	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
20	IOV18	bottom	power	Digital IO 1.8V Power Supply

21	SV11	bottom	power	SPAD Array 1.1V Power Supply
22	SVSS11	bottom	ground	SPAD Array 1.1V GND
23	VOP	bottom	power	SPAD Anode negative high voltage supply Vop
24	SV33	bottom	power	SPAD Array Vex power supply
25	SVSS33	bottom	ground	SPAD Array Vex GND
26	DV11	bottom	power	Digital 1.1V Power Supply
27	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
28	DV11	bottom	power	Digital 1.1V Power Supply
29	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
30	IOV18	bottom	power	Digital IO 1.8V Power Supply
31	SV11	bottom	power	SPAD Array 1.1V Power Supply
32	SVSS11	bottom	ground	SPAD Array 1.1V GND
33	VOP	bottom	power	SPAD Anode negative high voltage supply Vop
34	SV33	bottom	power	SPAD Array Vex power supply
35	SVSS33	bottom	ground	SPAD Array Vex GND
36	DV11	bottom	power	Digital 1.1V Power Supply
37	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
38	DV11	bottom	power	Digital 1.1V Power Supply
39	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
40	IOV18	bottom	power	Digital IO 1.8V Power Supply
41	SV11	bottom	power	SPAD Array 1.1V Power Supply
42	SVSS11	bottom	ground	SPAD Array 1.1V GND
43	VOP	bottom	power	SPAD Anode negative high voltage supply Vop
44	SV33	bottom	power	SPAD Array Vex power supply
45	SVSS33	bottom	ground	SPAD Array Vex GND
46	DV11	bottom	power	Digital 1.1V Power Supply
47	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
48	NC	bottom	I/O	Not Connected
49	NC	bottom	I/O	Not Connected
50	DV11	bottom	power	Digital 1.1V Power Supply
51	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
52	IOV18	bottom	power	Digital IO 1.8V Power Supply
53	SV11	bottom	power	SPAD Array 1.1V Power Supply
54	SVSS11	bottom	ground	SPAD Array 1.1V GND
55	VOP	bottom	power	SPAD Anode negative high voltage supply Vop
56	SV33	bottom	power	SPAD Array Vex power supply
57	SVSS33	bottom	ground	SPAD Array Vex GND

58	DV11	bottom	power	Digital 1.1V Power Supply
59	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
60	NC	bottom	I/O	Not Connected
61	NC	bottom	I/O	Not Connected
62	DV11	bottom	power	Digital 1.1V Power Supply
63	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
64	IOV18	bottom	power	Digital IO 1.8V Power Supply
65	NC	bottom	I/O	Not Connected
66	NC	bottom	I/O	Not Connected
67	DV11	bottom	power	Digital 1.1V Power Supply
68	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
69	NC	bottom	I/O	Not Connected
70	NC	bottom	I/O	Not Connected
71	NC	bottom	I/O	Not Connected
72	DV11	bottom	power	Digital 1.1V Power Supply
73	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
74	IOV18	bottom	power	Digital IO 1.8V Power Supply
75	NC	bottom	I/O	Not Connected
76	NC	bottom	I/O	Not Connected
77	TREFCLK	bottom	I/O	PLL Monitor Test RefClk input for function safety
78	DV11	bottom	power	Digital 1.1V Power Supply
79	DVSS	bottom	ground	Digital GND (1.1V and 1.8V)
80	ADR0	bottom	input	I2C Slave Address Select Signal 0
81	ADR1	right	input	I2C Slave Address Select Signal 1
82	ADR2	right	input	I2C Slave Address Select Signal 2
83	DV11	right	power	Digital 1.1V Power Supply
84	DVSS	right	ground	Digital GND (1.1V and 1.8V)
85	IOV18	right	power	Digital IO 1.8V Power Supply
86	SCL	right	input	I2C SCL or SPI SCLK signal
87	SDA	right	I/O	I2C SDA or SPI MOSI signal
88	SDO	right	output	SPI MISO signal
89	SSN	right	input	SPI CSN signal
90	DV11	right	power	Digital 1.1V Power Supply
91	DVSS	right	ground	Digital GND (1.1V and 1.8V)
92	IOV18	right	power	Digital IO 1.8V Power Supply
93	SV11	right	power	SPAD Array 1.1V Power Supply
94	SVSS11	right	ground	SPAD Array 1.1V GND

95	VOP	right	power	SPAD Anode negative high voltage supply Vop
96	SV33	right	power	SPAD Array Vex power supply
97	SVSS33	right	ground	SPAD Array Vex GND
98	CSEL	right	input	I2C/SPI interface selection (low: I2C, high: SPI)
99	SINT	right	output	Function Safety Interrupt Signal, high valid
100	DV11	right	power	Digital 1.1V Power Supply
101	DVSS	right	ground	Digital GND (1.1V and 1.8V)
102	IOV18	right	power	Digital IO 1.8V Power Supply
103	XCLK	right	input	I2C reference clock input
104	SV11	right	power	SPAD Array 1.1V Power Supply
105	SVSS11	right	ground	SPAD Array 1.1V GND
106	VOP	right	power	SPAD Anode negative high voltage supply Vop
107	SV33	right	power	SPAD Array Vex power supply
108	SVSS33	right	ground	SPAD Array Vex GND
109	MP_GND11A	right	ground	MIPI Analog 1.1V GND
110	MP_VCC18A	right	power	MIPI Analog 1.8V Power Supply
111	MP_VCC11A	right	power	MIPI Analog 1.1V Power Supply
112	MP_GND11A	right	ground	MIPI Analog 1.1V GND
113	MP_VCC11A	right	power	MIPI Analog 1.1V Power Supply
114	MP_GND11A	right	ground	MIPI Analog 1.1V GND
115	MP_VCC11A	right	power	MIPI Analog 1.1V Power Supply
116	MP_VCC12A	right	output	MIPI LDO output, connect to 2.2uF capacitor
117	MP_RBIA5	right	power	MIPI RBIA5 output, connect to 2K ohm resistance
118	MP_DP3	right	output	MIPI data lane 3
119	MP_DN3	right	output	MIPI data lane 3
120	MP_DP2	right	output	MIPI data lane 2
121	MP_DN2	right	output	MIPI data lane 2
122	MP_CKP	right	output	MIPI clock lane
123	MP_CKN	right	output	MIPI clock lane
124	MP_DP1	right	output	MIPI data lane 1
125	MP_DN1	right	output	MIPI data lane 1
126	MP_DP0	right	output	MIPI data lane 0
127	MP_DN0	right	output	MIPI data lane 0
128	SV11	right	power	SPAD Array 1.1V Power Supply
129	SVSS11	right	ground	SPAD Array 1.1V GND
130	VOP	right	power	SPAD Anode negative high voltage supply Vop
131	SV33	right	power	SPAD Array Vex power supply

132	SVSS33	right	ground	SPAD Array Vex GND
133	DV11	right	power	Digital 1.1V Power Supply
134	DVSS	right	ground	Digital GND (1.1V and 1.8V)
135	IOV18	right	power	Digital IO 1.8V Power Supply
136	TRIG_I	right	input	External VCSEL Emission Trigger Input signal
137	FSYNC	right	I/O	Image Frame Synchronization signal
138	SSYNC	right	I/O	Sub-Frame Synchronization signal
139	SV11	right	power	SPAD Array 1.1V Power Supply
140	SVSS11	right	ground	SPAD Array 1.1V GND
141	VOP	right	power	SPAD Anode negative high voltage supply Vop
142	SV33	right	power	SPAD Array Vex power supply
143	SVSS33	right	ground	SPAD Array Vex GND
144	DV11	right	power	Digital 1.1V Power Supply
145	DVSS	right	ground	Digital GND (1.1V and 1.8V)
146	IOV18	right	power	Digital IO 1.8V Power Supply
147	RSTB	right	input	Chip Reset signal, low valid
148	SWACK	right	input	VCSEL Driver Switch ACK signal, 1.8V power domain
149	SWREQ	right	output	VCSEL Driver Switch REQ signal, 1.8V power domain
150	DRVE	right	input	VCSEL Driver Error input signal, 1.8V power domain
151	DV11	right	power	Digital 1.1V Power Supply
152	DVSS	right	ground	Digital GND (1.1V and 1.8V)
153	IOV18	right	power	Digital IO 1.8V Power Supply
154	NC	right	input	Not Connected
155	NC	right	input	Not Connected
156	NC	top	input	Not Connected
157	DVSS	top	ground	Digital GND (1.1V and 1.8V)
158	DV11	top	power	Digital 1.1V Power Supply
159	NC	top	I/O	Not Connected
160	NC	top	I/O	Not Connected
161	NC	top	I/O	Not Connected
162	IOV18	top	power	Digital IO 1.8V Power Supply
163	DVSS	top	ground	Digital GND (1.1V and 1.8V)
164	DV11	top	power	Digital 1.1V Power Supply
165	VROLL4	top	I/O	Vertical Rolling Number signal 4, 1.8V power domain
166	VROLL3	top	I/O	Vertical Rolling Number signal 3, 1.8V power domain
167	VROLL2	top	I/O	Vertical Rolling Number signal 2, 1.8V power domain
168	DVSS	top	ground	Digital GND (1.1V and 1.8V)

169	DV11	top	power	Digital 1.1V Power Supply
170	VROLL1	top	I/O	Vertical Rolling Number signal 1, 1.8V power domain
171	VROLL0	top	I/O	Vertical Rolling Number signal 0, 1.8V power domain
172	IOV18	top	power	Digital IO 1.8V Power Supply
173	DVSS	top	ground	Digital GND (1.1V and 1.8V)
174	DV11	top	power	Digital 1.1V Power Supply
175	HROLL3	top	I/O	Horizontal Rolling Number signal 3, 1.8V power domain
176	HROLL2	top	I/O	Horizontal Rolling Number signal 2, 1.8V power domain
177	DVSS	top	ground	Digital GND (1.1V and 1.8V)
178	DV11	top	power	Digital 1.1V Power Supply
179	SVSS33	top	ground	SPAD Array Vex GND
180	SV33	top	power	SPAD Array Vex power supply
181	VOP	top	power	SPAD Anode negative high voltage supply Vop
182	SVSS11	top	ground	SPAD Array 1.1V GND
183	SV11	top	power	SPAD Array 1.1V Power Supply
184	IOV18	top	power	Digital IO 1.8V Power Supply
185	DVSS	top	ground	Digital GND (1.1V and 1.8V)
186	DV11	top	power	Digital 1.1V Power Supply
187	HROLL1	top	I/O	Horizontal Rolling Number signal 1, 1.8V power domain
188	HROLL0	top	I/O	Horizontal Rolling Number signal 0, 1.8V power domain
189	DVSS	top	ground	Digital GND (1.1V and 1.8V)
190	DV11	top	power	Digital 1.1V Power Supply
191	SVSS33	top	ground	SPAD Array Vex GND
192	SV33	top	power	SPAD Array Vex power supply
193	VOP	top	power	SPAD Anode negative high voltage supply Vop
194	SVSS11	top	ground	SPAD Array 1.1V GND
195	SV11	top	power	SPAD Array 1.1V Power Supply
196	IOV18	top	power	Digital IO 1.8V Power Supply
197	DVSS	top	ground	Digital GND (1.1V and 1.8V)
198	DV11	top	power	Digital 1.1V Power Supply
199	DVSS	top	ground	Digital GND (1.1V and 1.8V)
200	DV11	top	power	Digital 1.1V Power Supply
201	SVSS33	top	ground	SPAD Array Vex GND
202	SV33	top	power	SPAD Array Vex power supply
203	VOP	top	power	SPAD Anode negative high voltage supply Vop
204	SVSS11	top	ground	SPAD Array 1.1V GND
205	SV11	top	power	SPAD Array 1.1V Power Supply

206	IOV18	top	power	Digital IO 1.8V Power Supply
207	DVSS	top	ground	Digital GND (1.1V and 1.8V)
208	DV11	top	power	Digital 1.1V Power Supply
209	DVSS	top	ground	Digital GND (1.1V and 1.8V)
210	DV11	top	power	Digital 1.1V Power Supply
211	SVSS33	top	ground	SPAD Array Vex GND
212	SV33	top	power	SPAD Array Vex power supply
213	VOP	top	power	SPAD Anode negative high voltage supply Vop
214	SVSS11	top	ground	SPAD Array 1.1V GND
215	SV11	top	power	SPAD Array 1.1V Power Supply
216	IOV18	top	power	Digital IO 1.8V Power Supply
217	DVSS	top	ground	Digital GND (1.1V and 1.8V)
218	DV11	top	power	Digital 1.1V Power Supply
219	NC	top	I/O	Not Connected
220	NC	top	I/O	Not Connected
221	DVSS	top	ground	Digital GND (1.1V and 1.8V)
222	DV11	top	power	Digital 1.1V Power Supply
223	SVSS33	top	ground	SPAD Array Vex GND
224	SV33	top	power	SPAD Array Vex power supply
225	VOP	top	power	SPAD Anode negative high voltage supply Vop
226	SVSS11	top	ground	SPAD Array 1.1V GND
227	SV11	top	power	SPAD Array 1.1V Power Supply
228	IOV33	top	power	Digital IO 3.3V Power Supply
229	DVSS	top	ground	Digital GND (1.1V and 1.8V)
230	DV11	top	power	Digital 1.1V Power Supply
231	SWTREQ_V33	top	output	VCSEL Driver Switch REQ signal, 3.3V power domain
232	SWTACK_V33	top	input	VCSEL Driver Switch ACK signal, 3.3V power domain
233	IOV33	top	power	Digital IO 3.3V Power Supply
234	DVSS	top	ground	Digital GND (1.1V and 3.3V)
235	DV11	top	power	Digital 1.1V Power Supply
236	HROLLO_V33	top	output	Horizontal Rolling Number signal 0, 3.3V power domain
237	HROLL1_V33	top	output	Horizontal Rolling Number signal 1, 3.3V power domain
238	HROLL2_V33	top	output	Horizontal Rolling Number signal 2, 3.3V power domain
239	DVSS	top	ground	Digital GND (1.1V and 3.3V)
240	DV11	top	power	Digital 1.1V Power Supply
241	HROLL3_V33	left	output	Horizontal Rolling Number signal 3, 3.3V power domain
242	DRV_ERR_V33	left	input	VCSEL Driver Error input signal, 3.3V power domain

243	VROLL0_V33	left	output	Vertical Rolling Number signal 0, 3.3V power domain
244	IOV33	left	power	Digital IO 3.3V Power Supply
245	DVSS	left	ground	Digital GND (1.1V and 3.3V)
246	DV11	left	power	Digital 1.1V Power Supply
247	VROLL1_V33	left	output	Vertical Rolling Number signal 1, 3.3V power domain
248	VROLL2_V33	left	output	Vertical Rolling Number signal 2, 3.3V power domain
249	VROLL3_V33	left	output	Vertical Rolling Number signal 3, 3.3V power domain
250	VROLL4_V33	left	output	Vertical Rolling Number signal 4, 3.3V power domain
251	DV11	left	power	Digital 1.1V Power Supply
252	DVSS	left	ground	Digital GND (1.1V and 3.3V)
253	IOV33	left	power	Digital IO 3.3V Power Supply
254	SVSS33	left	ground	SPAD Array Vex GND
255	SV33	left	power	SPAD Array Vex power supply
256	VOP	left	power	SPAD Anode negative high voltage supply Vop
257	SVSS11	left	ground	SPAD Array 1.1V GND
258	SV11	left	power	SPAD Array 1.1V Power Supply
259	NC	left	output	Not Connected
260	AVSS33	left	ground	Analog 3.3V GND
261	NC	left	output	Not Connected
262	NC	left	output	Not Connected
263	VDDQ	left	input	EFUSE burning VDDQ, connect to GND
264	ADC_VREF	left	output	ADC VREF, connect to 1uF capacitor
265	ADC_VIN[1]	left	input	ADC VIN channel 1
266	ADC_VIN[0]	left	input	ADC VIN channel 0
267	AV33	left	power	Analog 3.3V Power Supply
268	AVSS33	left	ground	Analog 3.3V GND
269	NC	left	output	Not Connected
270	NC	left	output	Not Connected
271	NC	left	output	Not Connected
272	NC	left	output	Not Connected
273	PLL_CLKIN	left	input	PLL reference clock input, can use the same source with XCLK
274	AV33	left	power	Analog 3.3V Power Supply
275	SVSS33	left	ground	Analog 3.3V GND
276	SV33	left	power	SPAD Array Vex power supply
277	VOP	left	power	SPAD Anode negative high voltage supply Vop
278	SVSS11	left	ground	SPAD Array 1.1V GND
279	SV11	left	power	SPAD Array 1.1V Power Supply

280	AVSS11	left	ground	SPAD Array 1.1V GND
281	AV11	left	power	Analog 1.1V Power Supply
282	IOV33	left	power	Digital IO 3.3V Power Supply
283	DVSS	left	ground	Digital GND (1.1V and 3.3V)
284	DV11	left	power	Digital 1.1V Power Supply
285	TRIG_O3	left	output	VCSEL Trigger Output Signal 3
286	TRIG_O2	left	output	VCSEL Trigger Output Signal 2
287	TRIG_O1	left	output	VCSEL Trigger Output Signal 1
288	TRIG_O0	left	output	VCSEL Trigger Output Signal 0
289	IOV33	left	power	Digital IO 3.3V Power Supply
290	DVSS	left	ground	Digital GND (1.1V and 3.3V)
291	DV11	left	power	Digital 1.1V Power Supply
292	TRIG_F3	left	input	VCSEL Trigger Feedback Signal 3
293	TRIG_F2	left	input	VCSEL Trigger Feedback Signal 2
294	TRIG_F1	left	input	VCSEL Trigger Feedback Signal 1
295	TRIG_F0	left	input	VCSEL Trigger Feedback Signal 0
296	AVSS11	left	ground	SPAD Array 1.1V GND
297	AV11	left	power	Analog 1.1V Power Supply
298	SVSS33	left	ground	Analog 3.3V GND
299	SV33	left	power	SPAD Array Vex power supply
300	VOP	left	power	SPAD Anode negative high voltage supply Vop
301	SVSS11	left	ground	SPAD Array 1.1V GND
302	SV11	left	power	SPAD Array 1.1V Power Supply
303	NC	left	output	Not Connected
304	NC	left	output	Not Connected
305	IOV18	left	power	Digital IO 1.8V Power Supply
306	DVSS	left	ground	Digital GND (1.1V and 1.8V)
307	DV11	left	power	Digital 1.1V Power Supply
308	NC	left	output	Not Connected
309	NC	left	output	Not Connected
310	TSPAD_OUT2	left	output	Test SPAD Output 2
311	TSPAD_OUT1	left	output	Test SPAD Output 1
312	IOV18	left	power	Digital IO 1.8V Power Supply
313	DVSS	left	ground	Digital GND (1.1V and 1.8V)
314	DV11	left	power	Digital 1.1V Power Supply

4.2 BGA pin list

- 210 pins iBGA package
- Please refer to the previous chapter for pin list and descriptions
- Please contact ADPAS's engineer for the detailed information

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	#
15	GND	DV11	GND	SWTREQ_V33	SV11	DV11	IOV18	VOP	SV33	GND	SV11	GND	DV11	IOV18	GND	15
14	GND	DV11	GND	SWTACK_V33	SV11	DV11	GND	VOP	SV33	GND	SV11	GND	DV11	NC	NC	14
13	GND	DRV_ER_V33	HROLL3_V33	HROLL0_V33	GND	NC	HROLL0	HROLL1	HROLL2	GND	VROLL2	VROLL3	NC	NC	DRVE	13
12	IOV33	VROLL0_V33	HROLL2_V33	HROLL1_V33	GND	NC	HROLL3	VROLL0	VROLL1	VROLL4	NC	NC	SWREQ	SWACK	RSTB	12
11	VROLL4_V33	VROLL3_V33	VROLL2_V33	VROLL1_V33	GND	GND	GND	GND	GND	GND	GND	SSYNC	GND	MP_DP0	MP_DN0	11
10	AV33	VDDQ	NC	NC	GND	GND	Chip Top View EPAD⁽¹⁾ (NC)			GND	GND	FSYNC	GND	MP_DP1	MP_DN1	10
9	ADC_VREF	ADC_VIN[0]	ADC_VIN[1]	NC	GND	GND				GND	GND	TRGI	GND	MP_CKP	MP_CKN	9
8	NC	NC	NC	GND	GND	GND				GND	GND	GND	GND	MP_DP2	MP_DN2	8
7	AV11	GND	TRIG_O3	PLL_CLKIN	GND	GND				GND	GND	XCLK	GND	MP_DP3	MP_DN3	7
6	TRIG_O0	TRIG_O1	TRIG_O2	TRIG_F3	GND	GND	GND	GND	GND	GND	GND	CSEL	SINT	MP_RBIA	MP_VCC12A	6
5	IOV33	TRIG_F0	TRIG_F1	TRIG_F2	GND	GND	GND	GND	GND	GND	GND	TREFCLK	GND	MP_VCC18A	MP_VCC11A	5
4	GND	GND	NC	NC	GND	NC	NC	NC	NC	NC	NC	NC	ADR2	SDO	SSN	4
3	NC	NC	NC	NC	GND	GND	NC	NC	NC	NC	NC	NC	ADR1	SDA	SCL	3
2	GND	DV11	GND	SV11	GND	SV33	VOP	GND	DV11	GND	SV11	GND	DV11	ADR0	IOV18	2
1	GND	DV11	GND	SV11	IOV18	SV33	VOP	GND	DV11	GND	SV11	GND	DV11	GND	GND	1
#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	#

Chip TOP View iBGA 210pin package(Pad_0.5mm/Pitch_0.8mm)

Power Pins:

DV11	SV11	AV11	IOV18	IOV33	SV33	VOP
		MP_VCC11A	MP_VCC18A	AV33		

Note: (1) EPAD is only used for Thermal radiation, NC recommended.

Figure 4-1 Pin list top view for the BGA package

5 Block Diagram

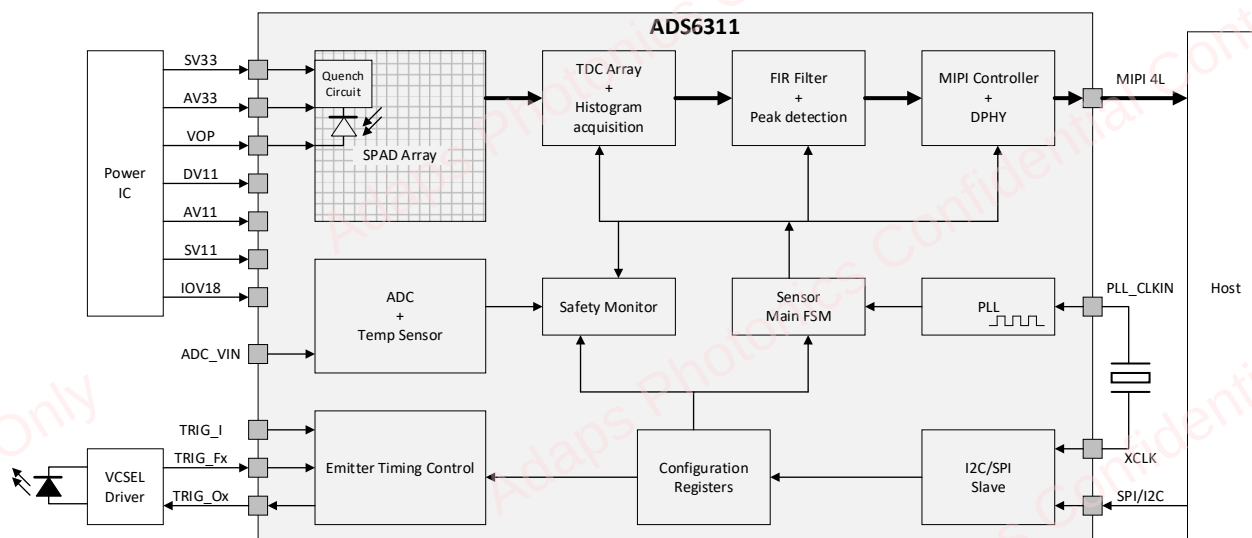


Figure 5-1 block diagram

6 Signal Processing Flow

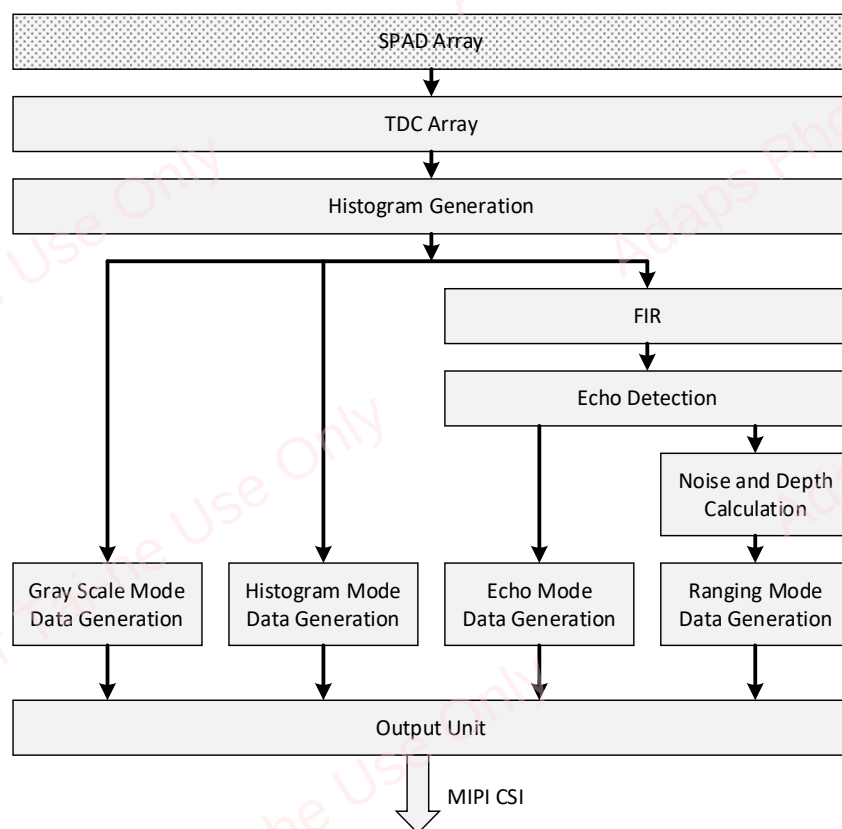


Figure 6-1 Signal Processing Flow

7 Electrical Characteristics

7.1 Absolute maximum ratings

Table 7-1 Absolute maximum ratings

Power description	Min.	Typ.	Max.	Unit
DV11, SV11, AV11, MP_VCC11A	-0.3	---	1.5	V
MP_VCC18A	-0.3	---	2.5	V
IOV18, IOV33	-0.3	---	4.0	V
AV33, SV33	-0.3	---	4.0	V
VOP	-24	---	0.3	V
Reflow (iBGA Package)			260	°C

7.2 Recommended Operating Conditions

Table 7-2 Recommended Operating Conditions

Power	GND	Power description	Min.	Typ.	Max.	Unit
DV11	DVSS	Digital core power	1.0	1.1	1.21	V
SV11	SVSS11	SPAD Array 1.1V Power	1.1	1.18	1.21	V
AV11	AVSS11	Analog core 1.1V power Supply	1.0	1.1	1.21	V
MP_VCC11A	MP_GND11A	MIPI 1.1V power Supply	1.0	1.1	1.21	V
IOV18	DVSS	Digital IO 1.8V power	1.7	1.8	1.98	V
MP_VCC18A	MP_GND11A	MIPI 1.8V Power Supply	1.7	1.8	1.98	V
IOV33	DVSS	Digital IO 3.3V Power	3.0	3.3	3.6	V
AV33	AVSS33	Analog core 3.3V power	3.0	3.3	3.6	V
SV33	SVSS33	SPAD excess bias voltage (Vex)	3.0	3.3	3.6	V
VOP	NA	SPAD Anode negative high voltage	-24	-20 ⁽¹⁾	-18	V

(1) Room temperature.

(2) Power supply combination suggestion

- a) SV11, DV11 and AV11 should be separated
- b) SV33 and AV33 should be separated
- c) IOV18, IOV33, MP_VCC11A, MP_VCC18A can share the same DVSS as ground

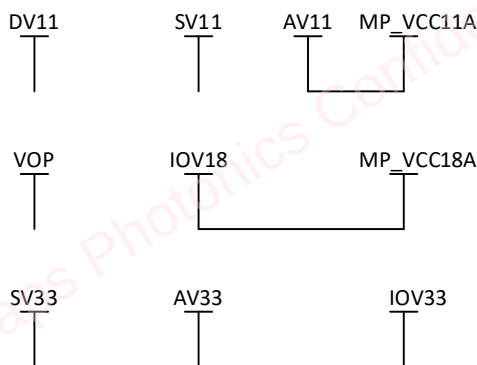


Figure 7-1 power supply combination suggestion

7.3 DC Characteristics

7.3.1 IOV18 digital input and output pins

Table 7-3 IOV18 digital IO DC characteristics

Item	Pin	Symbol	Min.	Typ.	Max.	Unit
Digital input voltage	RSTB, ADR0, ADR1, ADR2, FSYNC, SSYNC, XCLK, SCL, SDA, SSN, CSEL, SACK, TRIG_I, DRVE, TREFCLK	V_{IH}	$0.7 \cdot IOV18$	---	---	V
		V_{IL}	---	---	$0.3 \cdot IOV18$	V
Digital output voltage	FSYNC, SSYNC, SDA, SDO, SINT, SREQ,	V_{OH}	$0.7 \cdot IOV18$	---	---	V
		V_{OL}	---	---	0.4	V
Digital output current	VROLL4, VROLL3, VROLL2, VROLL1, VROLL0, HROLL3, HROLL2, HROLL1, HROLL0	I_{OH}	1.5	$6^{(1)}$	6	mA
		I_{OL}	1.5	$6^{(1)}$	6	mA
Internal Pull-Up resistor	---	R_{PU}	50	179.6	400	k Ω
Internal Pull-Down resistor	---	R_{PD}	50	154.1	400	k Ω

(1) The driving capacity is fixed at 6mA.

7.3.2 IOV33 digital input and output pins

Table 7-4 IOV33 digital IO DC characteristics

Item	Pin	Symbol	Min.	Typ.	Max.	Unit
Digital input voltage	SACK_V33, DRVE_V33,	V_{IH}	$0.7 \cdot IOV33$	---	---	V

	TRIG_F0, TRIG_F1, TRIG_F2, TRIG_F3	V _{IL}	---	---	0.3*IOV33	V
Digital output voltage	SREQ_V33, VROLL4_V33, VROLL3_V33, VROLL2_V33, VROLL1_V33, VROLL0_V33, HROLL3_V33, HROLL2_V33, HROLL1_V33, HROLL0_V33, TRIG_O0, TRIG_O1, TRIG_O2, TRIG_O3	V _{OH}	0.7*IOV33	---	---	V
		V _{OL}	---	---	0.4	V
Digital output current		I _{OH}	4	4 ⁽¹⁾ 8 ⁽²⁾	16	mA
		I _{OL}	4	4 ⁽¹⁾ 8 ⁽²⁾	16	mA
Internal Pull-Up resistor	---	R _{PU}	40	75	190	kΩ
Internal Pull-Down resistor	---	R _{PD}	30	75	190	kΩ

(1) The driving capacity is fixed at 4mA except for TRIG_O0, TRIG_O1, TRIG_O2, TRIG_O3.

(2) The optimal driving capacity is 8mA for TRIG_O0, TRIG_O1, TRIG_O2, TRIG_O3.

7.4 Current characteristics

7.4.1 Standby current

After all power supplies ready and RSTB released, the chip goes into HW standby.

Table 7-5 Hardware standby Current characteristics

Symbol	Item	Min.	Typ. ⁽¹⁾	Max ⁽²⁾	Unit
I _{DV11_STB}	Standby current into DV11 power lines	---	30	800	mA
I _{SV11_STB}	Standby current into SV11 power lines	---	5	200	mA
I _{MP_VCC11A_STB}	Standby current into MP_VCC11A power lines	---	0.05	1	mA
I _{MP_VCC18A_STB}	Standby current into MP_VCC18A power lines	---	0.02	0.1	mA
I _{AV11_STB}	Standby current into AV11 power lines	---	0.1	1	mA
I _{IOV18_STB}	Standby current into IOV18 power lines	---	0.1	0.5	mA
I _{IOV33_STB}	Standby current into IOV33 power lines	---	1	2	mA
I _{AV33_STB}	Standby current into AV33 power lines	---	0.6	1	mA
I _{SV33_STB}	Standby current into SV33 power lines	---	0.5	1	mA
I _{VOP_STB}	Standby current into VOP power lines	---	0.1	1	mA
P _{total}	Total power in HW standby	---	47	1135	mW

(1) Ta=Room temperature

(2) Based on Characterization @Ta=105°C

7.4.2 Operating average current

Table 7-6 Operating average current characteristics

Symbol	Item	Min.	Typ. ⁽¹⁾	Typ. ⁽²⁾	Typ. ⁽³⁾	Unit
I _{DV11_AVE}	Average current into DV11 power lines	---	660	1500	2100	mA
I _{SV11_AVE}	Average current into SV11 power lines	---	40	300	400	mA
I _{MP_VCC11A_AVE}	Average current into MP_VCC11A power lines	---	13	13	14	mA
I _{MP_VCC18A_AVE}	Average current into MP_VCC18A power lines	---	1	1	1	mA
I _{AV11_AVE}	Average current into AV11 power lines	---	5	5	8	mA
I _{IOV18_AVE}	Average current into IOV18 power lines	---	1	1	1	mA
I _{IOV33_AVE}	Average current into IOV33 power lines	---	2	2	3	mA
I _{AV33_AVE}	Average current into AV33 power lines	---	2	2	4	mA
I _{SV33_AVE}	Average current into SV33 power lines	---	7	50	75	mA
I _{VOP_AVE}	Average current into VOP power lines	---	7	50	75	mA
P _{total}	Total power in every rolling	---	975	3405	4595	mW

Note:

The test condition of Typical scenario is as below, the value is based on characterization for reference.:

Note	Scenario	Ambient Temp (°C)	DT(ns)	SPAD Trigger Rate per SPAD(Mcps) *	VOP(V)	Work Mode	Exposure Time(us)	IDLE Time(us)
(1)	Typ. ⁽¹⁾	25	10	10	-19.7	ECHO	1000	1000
(2)	Typ. ⁽²⁾	25	10	50	-19.7	Ranging	1000	500
(3)	Typ. ⁽³⁾	125	10	50	-22.0	Ranging	1000	500

*The SPAD trigger rate calculation formula: XX Mcps=bin counts/shot number/bin width/9

7.4.3 Operating peak current

Table 7-7 Operating peak current characteristics

Symbol	Item	Min.	Typ.	Max. ⁽¹⁾	Unit
I _{DV11_PEAK}	Peak current into DV11 power lines	---	---	4600	mA
I _{SV11_PEAK}	Peak current into SV11 power lines	---	---	1200	mA
I _{MP_VCC11A_PEAK}	Peak current into MP_VCC11A power lines	---	---	16	mA
I _{MP_VCC18A_PEAK}	Peak current into MP_VCC18A power lines	---	---	3	mA
I _{AV11_PEAK}	Peak current into AV11 power lines	---	---	6	mA
I _{IOV18_PEAK}	Peak current into IOV18 power lines	---	---	48 ⁽²⁾	mA

I _{IOV33_PEAK}	Peak current into IOV33 power lines	---	---	64 ⁽³⁾	mA
I _{AV33_PEAK}	Peak current into AV33 power lines	---	---	5	mA
I _{SV33_PEAK}	Peak current into SV33 power lines	---	---	180	mA
I _{VOP_PEAK}	Peak current into VOP power lines	---	---	180	mA

(1) Ta=125°C, all power supplies with capacitors

(2) when VROLL and HROLL pins toggling simultaneous.

(3) when 4 TRIG_Ox pins asserted simultaneous.

Note: Based on characterization for reference.

7.5 AC Characteristics

Table 7-8 AC characteristic

Symbol	Item	Min.	Typ.	Max.	Unit	Remark
f _{XCLK}	XCLK frequency	20	24	30	MHz	
f _{PLL_CLKIN}	PLL_CLKIN frequency					
f _{SPI}	SPI SCK frequency	---	10	30	MHz	
f _{I2C}	I2C SCL frequency	---	1000	1000	KHz	
t _{SPI_SU}	SPI Setup time (Start condition)	0.5	---	---	1/ f _{SPI}	
t _{I2C_SU}	I2C Setup time (Start condition)	1.1	---	---	1/ f _{XCLK}	
t _{RSTB_LW}	RSTB Low Level Pulse Width	1200	---	---	ns	

7.5.1 TRIG_O signals (TRIG_O0/1/2/3)

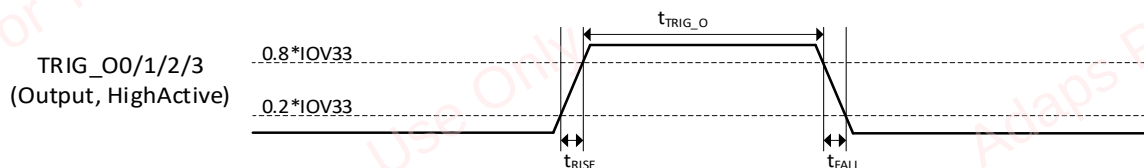


Figure 7-2 TRIG_O0/1/2/3 Waveform schematic diagram

Table 7-9 TRIG_O output characteristics

Symbol	Item	Min.	Typ.	Max.	Unit	Remark
t _{RISE}	TRIG_O signals rise time	---	0.8	---	ns	0.2*IOV33 to 0.8*IOV33
t _{FALL}	TRIG_O signals fall time	---	0.8	---	ns	0.8*IOV33 to 0.2*IOV33
t _{TRIG_O}	TRIG_O signals pulse width	2	4	16384	ns	Min Step: 2ns

7.5.2 TRIG_I signal

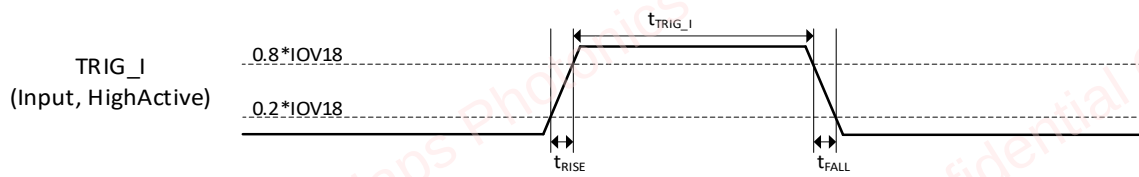


Figure 7-3 TRIG_I Waveform schematic diagram

Table 7-10 TRIG_I Input characteristics

Symbol	Item	Min.	Typ.	Max.	Unit	Remark
t_{RISE}	TRIG_I signal rise time	---	---	100	ns	0.2*IOV18 to 0.8*IOV18
t_{FALL}	TRIG_I signal fall time	---	---	100	ns	0.8*IOV18 to 0.2*IOV18
t_{TRIG_I}	TRIG_I signal pulse width	4	---	---	ns	Posedge detection

7.5.3 SPI timing

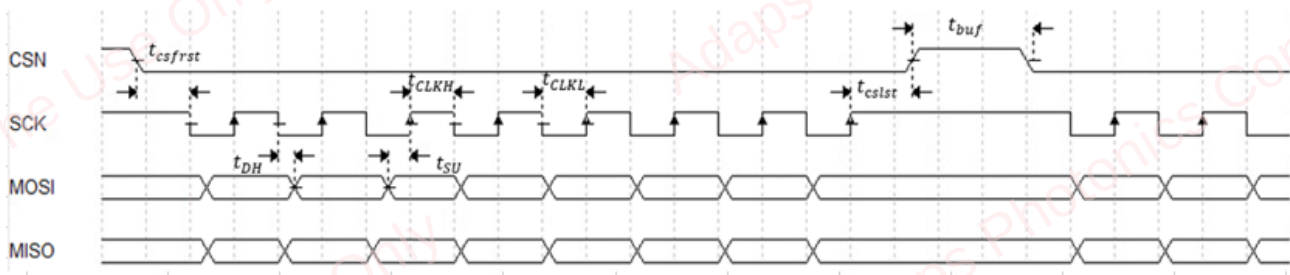


Figure 7-4 SPI timing diagram

Table 7-11 SPI timing characteristics

Symbol	Item	Unit	Min.	Max.
f_{SPI}	SPI SCK frequency	MHz	---	30
t_{csfrst}	SPI CSN asserted to first SCK edge	ns	0	---
t_{cslst}	SPI last SCK edge to SCK de-asserted	ns	0	---
t_{SU}	SPI data setup time	ns	0	---
t_{DH}	SPI data hold time	ns	0	---
t_{buf}	SPI bus idle time between two transfers	ns	4	---
t_{CLKL}	SPI SCK low time	ns	10	---
t_{CLKH}	SPI SCK high time	ns	10	---

7.5.4 I2C timing

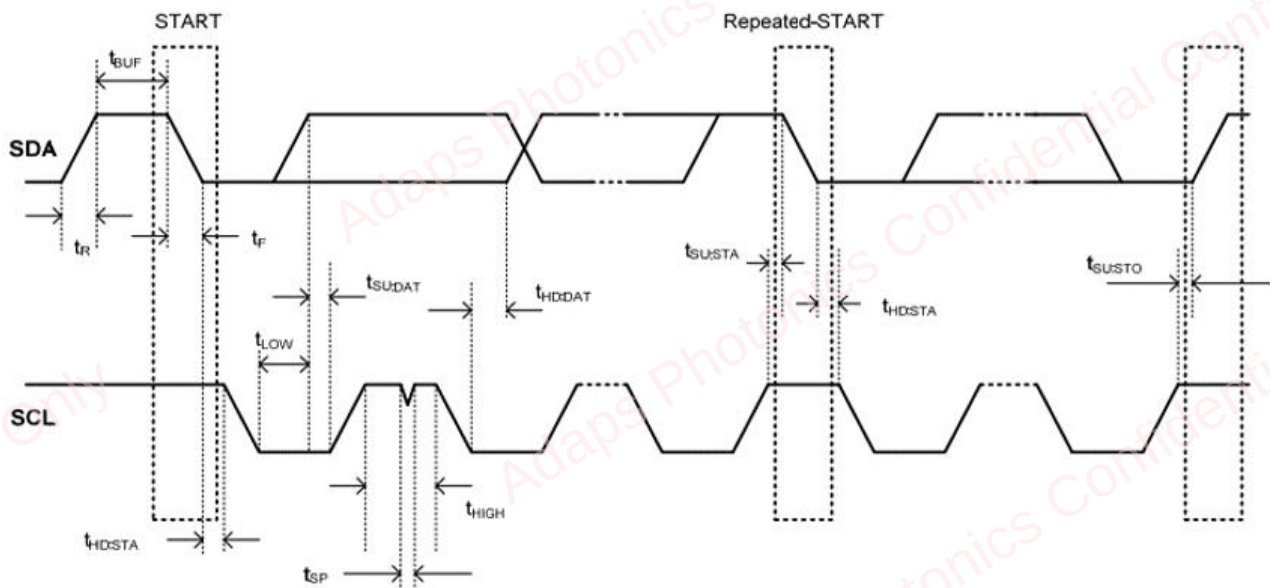


Figure 7-5 I2C timing diagram

Table 7-12 I2C timing characteristics

Symbol	Item	Std Mode		Fast Mode		FastPlus Mode		Unit
		Min	Max	Min	Max	Min	Max	
f_{I2C}	I2C SCL clock frequency	0	100	0	400	0	1000	kHz
t_{LOW}	I2C SCL low time	1	---	0.4	---	0.3	---	us
t_{HIGH}	I2C SCL high time	0.5	---	0.2	---	0.1	---	us
$t_{HD.STA}$	I2C hold time after (repeated) start condition. After this period, the first clock is generated.	0.03	---	0.03	---	0.04	---	us
$t_{SU.STA}$	I2C setup time for a repeated START condition	0.07	---	0.05	---	0.03	---	us
$t_{HD.DAT}$	I2C data in hold time	0.01	---	0.01	---	0.03	---	us
$t_{SU.DAT}$	I2C data in setup time	40	---	40	---	40	---	ns
$t_{SU.STO}$	I2C setup time for STOP condition	0	---	0	---	0	---	us
t_{BUF}	I2C bus free time between a STOP and START condition	0.09	---	0.08	---	0.08	---	us
t_{SP}	I2C time of spike suppression	0	80	0	80	0	80	ns
t_R	I2C rise time of both SDA and SCL signals	---	1000	---	300	---	120	ns
t_F	I2C fall time of both SDA and SCL signals	---	300	---	300	---	120	ns
duty	I2C SCL clock duty	5	90	5	85	5	70	%

7.6 SPAD Characteristics

Table 7-13 SPAD characteristics

Item	Symbol	Conditions	Characteristics			Unit	Remark
			Min.	Typ.	Max.		
Breakdown voltage	VBD	-40°C~125°C	-22.5	-19.7	-18.0	V	SPADs turn on > 10%
Breakdown voltage Temperature	VBD/Tj	-40°C~125°C	--	-23	--	mV/°C	
Dark count rate	DCR	25°C	--	300	1.5k	cps	The median value of DCR in the chip.
		105°C	--	300K	1M	cps	
Photon detection efficiency	PDE	-40°C~125°C	--	26	--	%	The median value of PDE in the chip.
Dead time (Pulse width)	DT_MEAN	-40°C~125°C	--	10*	--	ns	The mean value of DT, configurable.
	DT_FWHM	-40°C~125°C	--	--	4*	ns	FWHM of occurrence distribution of dead time in the specific test SPAD.
After-pulse probability	AP	-40°C~125°C	--	1*	--	%	
Crosstalk probability between SPAD	CT	-40°C~125°C	--	1*	--	%	The average value of adjacent SPAD.
SPAD output jitter	Jitter	-40°C~125°C	--	250*	--	ps	FWHM value @70ps pulse laser Test.
Incident angle characteristics	CRA	90% sensitivity relative to 0°	--	--	20	deg	

Note: Based on characterization. Table shows characteristic as shipped under the following voltage conditions.

VOP=Optimal voltage (VOV = 3.3V), Vex (SV33) =3.3V, laser wavelength (λ =940nm)

* These values tested on specific SPAD test pin, these characteristics are reference and not guaranteed.

7.7 Temperature sensor Characteristics

Table 7-14 Temperature sensor characteristics

Item	Min.	Typ.	Max.	Unit	Remark
Range	-40		125	°C	
Resolution		0.25		°C	
Accuracy		± 1	± 3	°C	

7.8 ADC Characteristics

Table 7-15 ADC characteristics

Item	Min.	Typ.	Max.	Unit	Remark
Resolution		10		bits	
ADC Reference Voltage	-	2.5		V	Internal Generation
Input Voltage ranges	0		ADC_VRef	V	
ADC clock frequency	1		16	MHz	Register configurable

8 Power up and power down sequence

8.1 Power up sequence

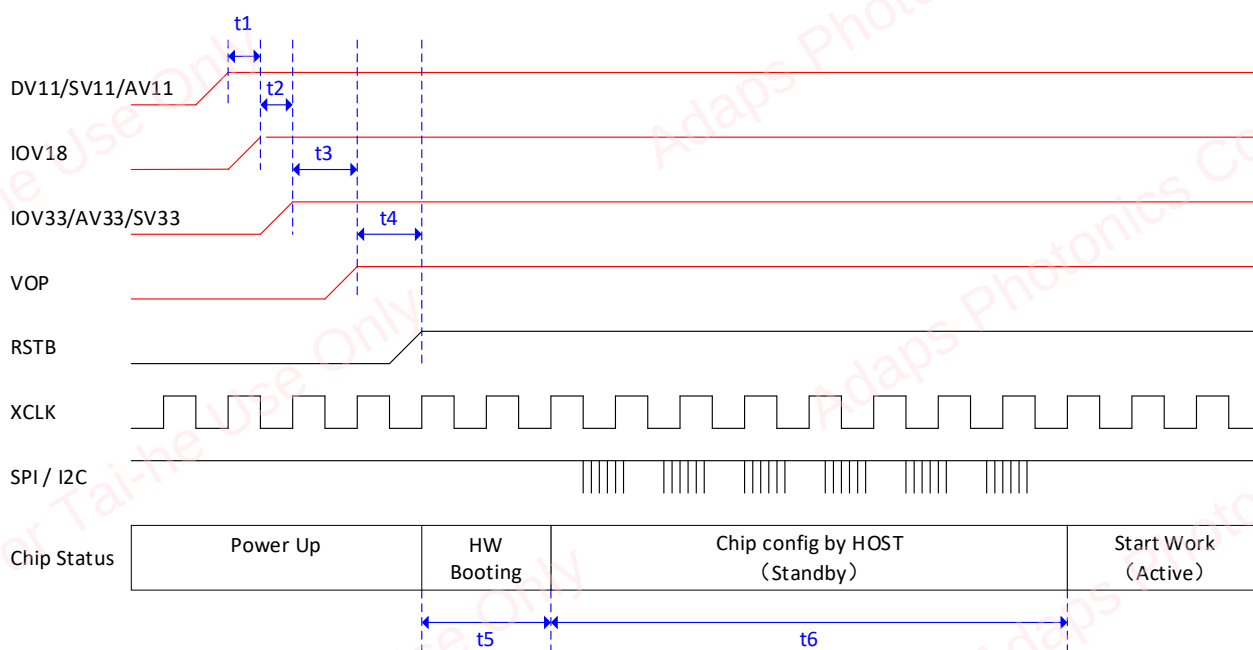


Figure 8-1 Power up sequence

The power up sequence should comply with the following principle: the lowest voltage power up at first and the highest voltage at the last, then release the RSTB. The timing constraints are listed as below.

Table 8-1 power up sequence timing constraints

Constraint	Label	Min.	Max.	Unit	Remark
1.1V power up to IO 1.8V power up	t1	100	---	us	
IO 1.8V power up to IOV33/AV33/SV33 power up	t2	100	---	us	

IOV3/AV33/SV33 power up to VOP power up	t3	100	---	us	VOP should be powered up slowly.
VOP power up to RSTB release	t4	100	---	us	
RSTB release to HW booting finished	t5	220	---	us	Depends on XCLK frequency
SW configuration duration	t6	2	---	ms	Depends on SPI/I2C speed and configuration contents

8.2 Power down sequence

The power down sequence is reversed with the power up sequence. It's recommended to tie low the RSTB pin at first, then power down the highest voltage, and the lowest voltage power down at the last.

9 Scanning mode

This chip support 1D(Dimension) and 2D scanning mode. Here are the 1D and 2D scanning mode definitions.

1D scan: The light strip is a whole one in the horizontal direction, and the rolling direction only moves in the vertical direction.

2D scan: The light strip is a smaller one than 1D scan, and it is divided into several segmentations in the horizontal direction. That's to say, the rolling direction not only moves in the vertical direction, but also the horizontal direction as well.

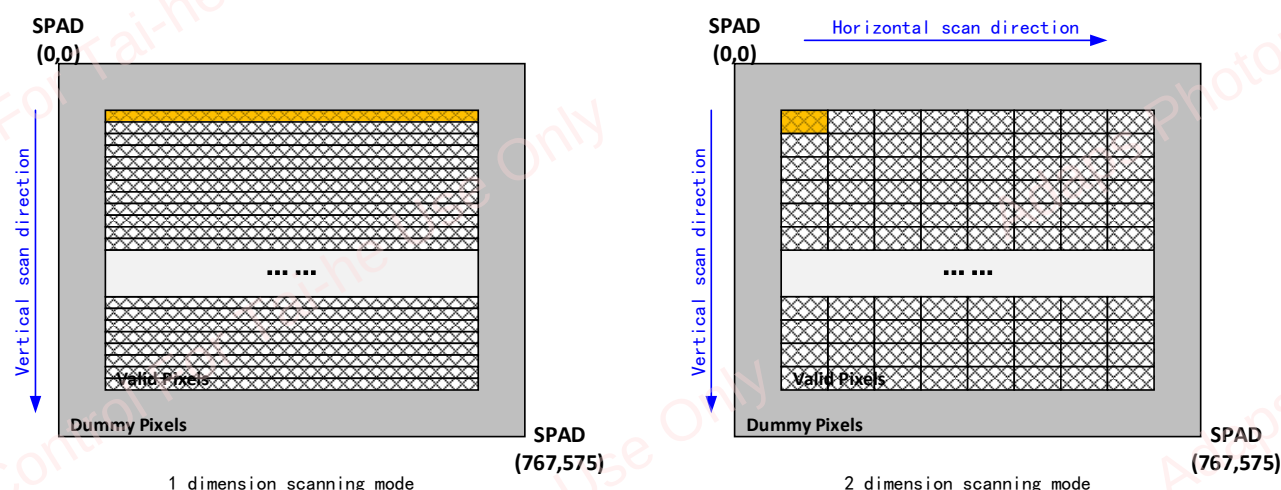


Figure 9-1 1D and 2D scanning mode diagram

Note: The yellow parts in the above diagram are the ROIs(Region Of Interest).

9.1 1D Scanning mode

9.1.1 Timing Control

Every image frame is started with a FSYNC pulse, and sub-frame is indicated by the SSYNC pulse. Every rolling will output one sub-frame. The VROLL and HROLL pins indicate the current rolling number.

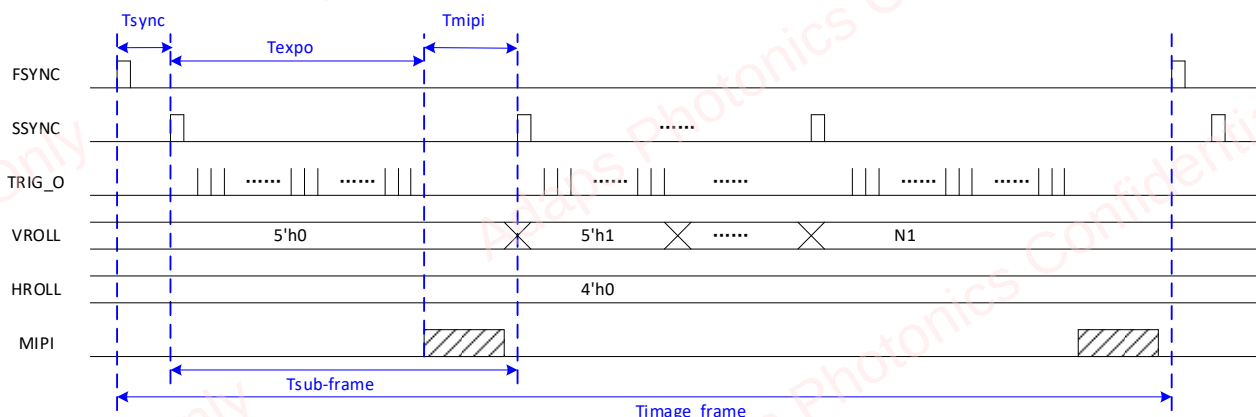


Figure 9-2 1D scanning mode timing

Note: The HROLL is a constants value(4'b0) in 1D scanning mode.

Table 9-1 The rolling timing constraints

Symbol	Item	Min.	Typ.	Max.	Unit
Tsync	FSYNC valid to SSYNC valid time	1	---	---	Tsysclk
Texpo	Exposure and histogram acquisition duration time	10	---	40960	us
Tmipi	MIPI data output time	400 ⁽¹⁾	---	---	us
Tsub-frame	Sub-frame duration time	420 ⁽²⁾	---	---	us
Timage_frame	Image-frame duration time(1/frame_rate)	---	---	---	us

(1): Depends on the ROI configuration and output mode. This ROI covers 256(H)*6(V) pixels, MIPI ranging output mode.

(2): Depends on the ROI configuration, exposure time and output mode. This ROI covers 256(H)*6(V) pixels. MIPI ranging output mode, and 10us exposure time.

9.1.2 ROI shapes

In the 1D scanning mode, up to 256(H)*6(V) pixels can be opened as ROI every rolling.

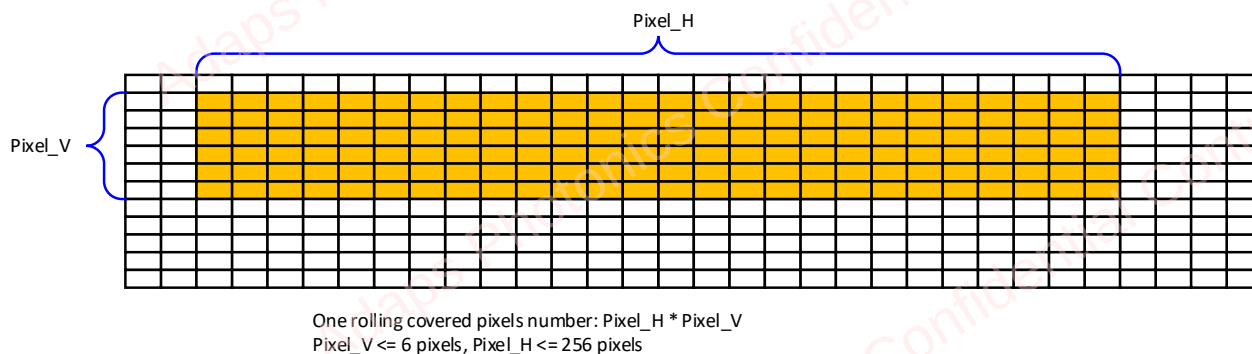


Figure 9-3 The basic ROI in 1D scanning mode

According to different ROI configurations, the distortion light strips are supported. For example, the declining light strap. But in this case, the VCSEL Tx light strips should cover more pixels than $256(H)*6(V)$.

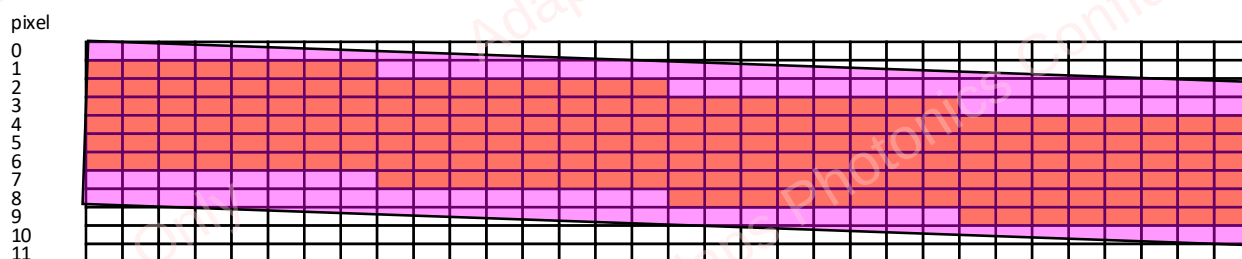


Figure 9-4 the ROI for declining light strip

9.2 2D Scanning mode

9.2.1 Timing control

The 2D scanning timing control is similar with 1D scan. The only difference is the HROLL pins are toggled every rolling.

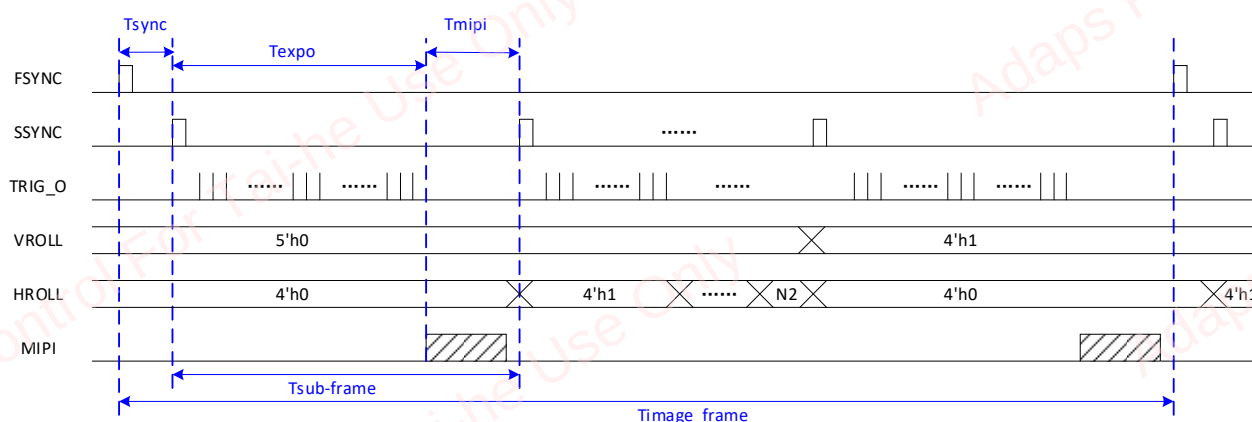


Figure 9-5 2D scanning mode timing

9.2.2 ROI shapes

The basic ROI shapes and rolling sequence in 2D scan mode is demonstrated below.

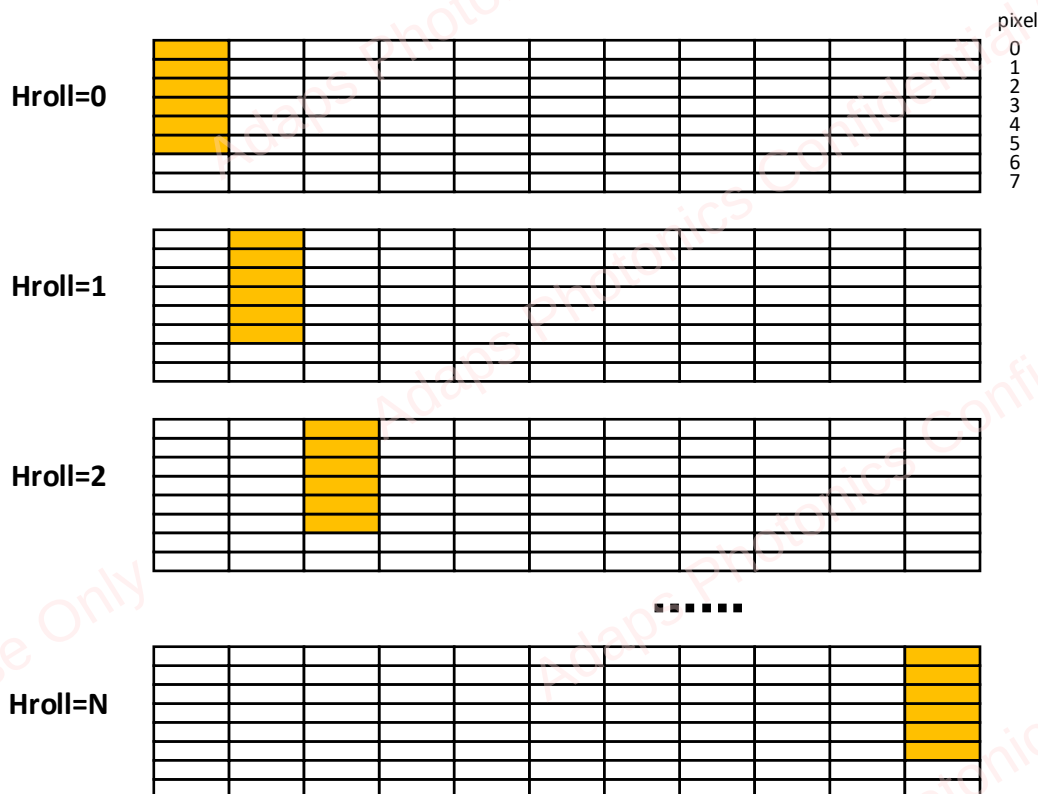


Figure 9-6 The basic ROI in 2D scanning mode

10 Communication interface

10.1 I2C interface

The chip can be controlled by an I2C bus. It supports 7bits device addressing and standard, fast mode(400Kbps) and fast mode plus mode(1Mbps). Read and Write transactions comply with the standard set by Philips (now NXP).

For a complete description of the I²C protocol, please review the NXP I²C design specification.

The I2C interface support 4 read/write operation:

- Single read
- Sequential read
- Single write

- Sequential write

10.1.1 Single Read

The HOST read data from a random address. The data format is as following.

S	Slave address[6:0]	0	A	Register address[15:8]	A	Register address[7:0]	A	Sr / P	Slave address[6:0]	1	A	Read data[7:0]	$\overline{A}$	P
---	--------------------	---	---	------------------------	---	-----------------------	---	--------	--------------------	---	---	----------------	----------------	---

10.1.2 Sequential Read

The HOST read data from the successive addresses, the data format is similar to the single read, the only difference is there are several read bytes before STOP asserted. The address will be incremented internally.

S	Slave address[6:0]	0	A	Register address[15:8]	A	Register address[7:0]	A	Sr / P	Slave address[6:0]	1	A	Read data[7:0]	A	Read data[7:0]	$\overline{A}$	P
---	--------------------	---	---	------------------------	---	-----------------------	---	--------	--------------------	---	---	----------------	---	----------------	----------------	---

N bytes of data, the register address incremented automatically

10.1.3 Single write

The HOST write data to a random address. The data format is as following.

S	Slave address[6:0]	0	A	Register address[15:8]	A	Register address[7:0]	A	Write data[7:0]	$\overline{A}$ / A	P
---	--------------------	---	---	------------------------	---	-----------------------	---	-----------------	--------------------	---

10.1.4 Sequential write

The HOST write data to the successive addresses, the data format is similar to the single write, the only difference is there are several write bytes before STOP asserted. The address will be incremented internally.

S	Slave address[6:0]	0	A	Register address[15:8]	A	Register address[7:0]	A	Write data[7:0]	A	Write data[7:0]	$\overline{A}$ / A	P
---	--------------------	---	---	------------------------	---	-----------------------	---	-----------------	---	-----------------	--------------------	---

N bytes of data, the register address incremented automatically

10.1.5 I2C device address

The default I2C device address depends on ADR2, ADR1 and ADR0 pad, there are eight configurations.

Table 10-1 I2C slave address configuration

ADR2	ADR1	ADR0	I2C_ADDRESS
0	0	0	7'h4A (default value)
0	0	1	7'h4B
0	1	0	7'h4C
0	1	1	7'h4D
1	0	0	7'h5A
1	0	1	7'h5B
1	1	0	7'h5C
1	1	1	7'h5D

Note: Those three pins are pulled down inside by default. That's to say, the I2C_ADDRESS is 7'h4A by default when ADR0~ADR3 are not connected.

10.2 SPI interface

The chip can be controlled by SPI interface as well, but I2C or SPI cannot communicate concurrently, only one protocol interface is supported. SPI interface has the higher speed but needs more pins compare with I2C interface. The SPI MODE0(CPOL=CPHA=0) and MODE3(CPOL=CPHA=1) are supported.

10.2.1 Single or sequential read

The HOST can read data from the random or successive addresses. The address will be incremented internally at successive mode.

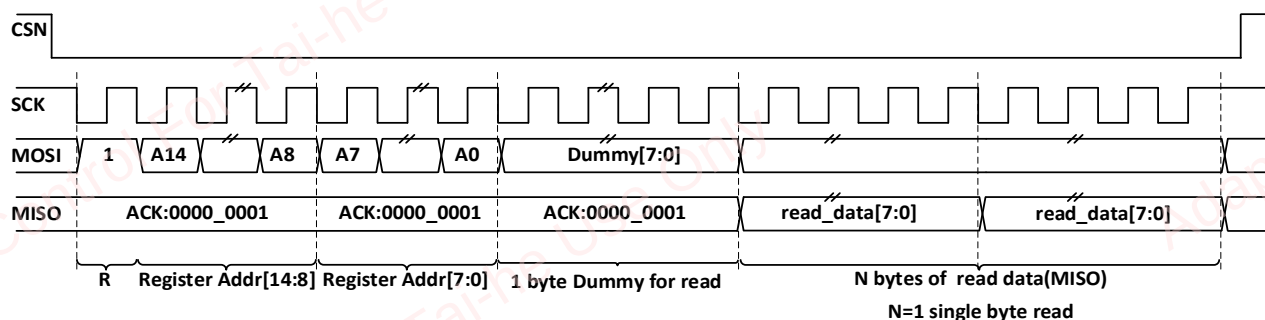
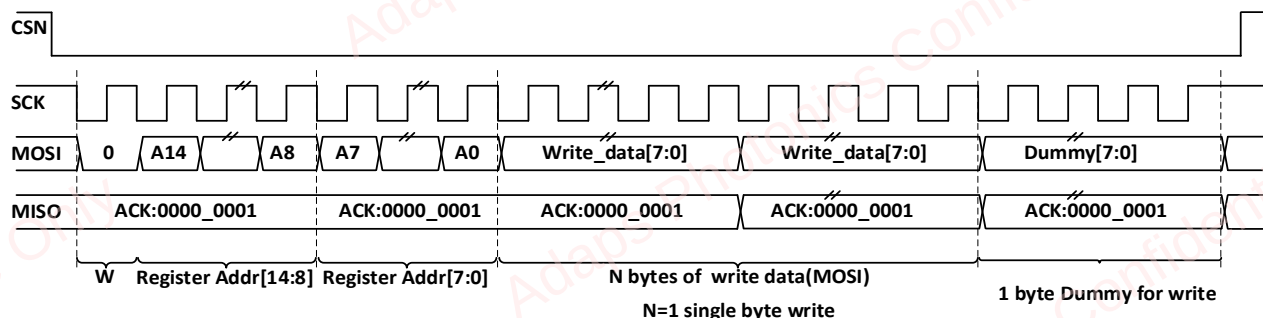


Figure 10-1 SPI read

Note: There is one dummy byte data after the address phase in read operation.

10.2.2 Single or sequential write

The HOST can write data from the random or successive addresses. The address will be incremented internally at successive mode.



Note: There is one dummy byte data after the valid write data phase in write operation, but this dummy byte data would not be written into the memory or register if the SCK stop toggling after this dummy byte.

10.3 MIPI interface

10.3.1 Frame types definition

To avoid ambiguity, here are the definitions of three MIPI frame types.

Data frame: The physical frame, it's related to MIPI VC (Virtual Channel) closely. One Virtual Channel corresponds to one data frame.

Sub-frame: It's composed by data frame(s) and related to Rolling Scan closely. One rolling shutter corresponds to one sub-frame. This chip, one sub-frame is composed by 2 data frames (VC0 and VC1 data frame).

Image frame: It's similar with the concept of CIS (CMOS Image Sensor) frame. Meanwhile, it's related to Frame Rate closely. For example, if the image frame rate was 30fps, that means every image frame output in 33.3 milliseconds. This chip, one image frame is composed by several sub-frames, the number of sub-frames depends on the rolling number.

The bellowing abbreviations will be used to demonstrate the MIPI frame format.

FS: Frame Star

FE: Frame End

PH: Package Head

PE: Package End

Frame Info: Additional Frame Information

V_ROLL_NUM: Vertical Rolling Number

H_ROLL_NUM: Horizontal Rolling Number

10.3.2 data frame format in gray scale mode

In gray scale mode, this chip outputs the intensity counts data of every pixel.



Figure 10-2 data frame format in gray scale mode

Note:

- 1) The sub-frame number K is equal to V_ROLL_NUM.
- 2) The packet number n depends on the ROI configuration.
- 3) There are two data types in one data frame (0x2C and 0x30).
- 4) On the MIPI bus, one packet of VC1 is transmitted first, then one packet of VC0 later, and the two VCs packet alternate back and forth.

10.3.3 data frame format in ranging mode

In ranging mode, this chip outputs the depth value data of every pixel.

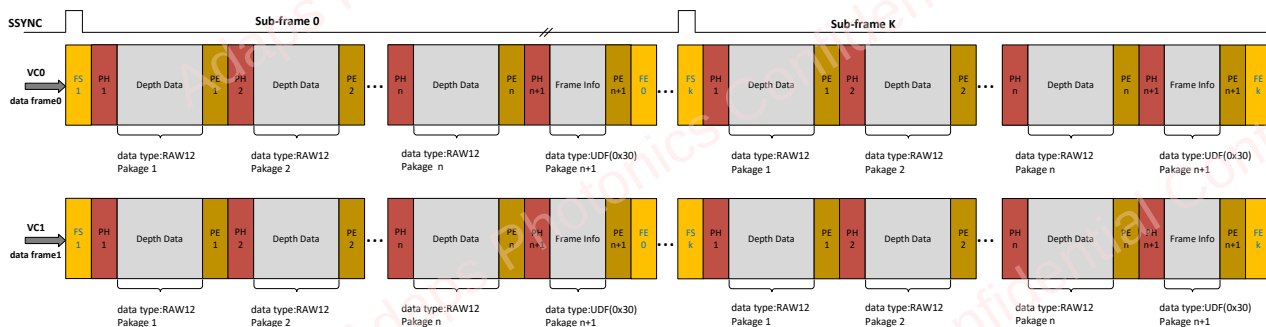


Figure 10-3 data frame format in ranging mode

Note:

- 1) The sub-frame number K is equal to $V_ROLL_NUM@1D$ scan or $V_ROLL_NUM * H_ROLL_NUM@2D$ scan.
- 2) The packet number n depends on the ROI configuration.
- 3) There are two data types in one data frame (0x2C and 0x30).
- 4) On the MIPI bus, one packet of VC1 is transmitted first, then one packet of VC0 later, and the two VCs packet alternate back and forth.

10.3.4 data frame format in echo mode

In echo mode, this chip outputs the echoes data of every pixel.

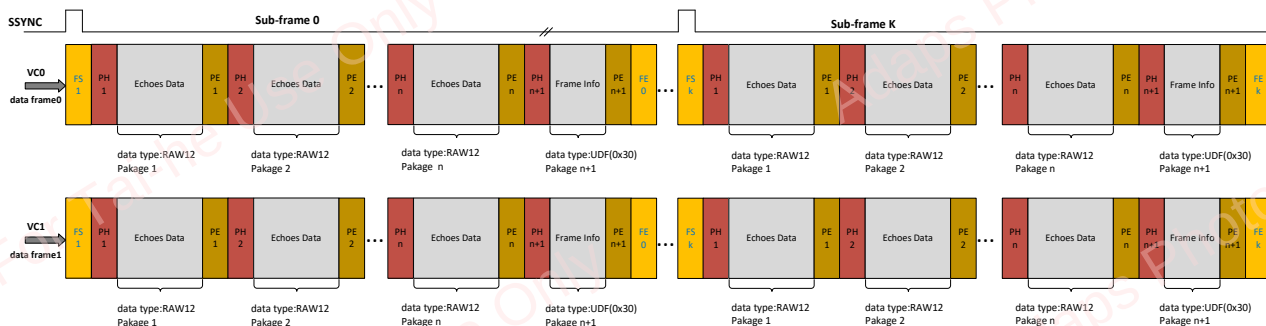


Figure 10-4 data frame format in echo mode

Note:

- 1) The sub-frame number K is equal to $V_ROLL_NUM@1D$ scan or $V_ROLL_NUM * H_ROLL_NUM@2D$ scan.
- 2) The packet number n depends on the ROI configuration.
- 3) There are two data types in one data frame (0x2C and 0x30).
- 4) On the MIPI bus, one packet of VC1 is transmitted first, then one packet of VC0 later, and the two VCs packet alternate back and forth.

10.3.5 data frame format in histogram mode

In histogram mode, this chip outputs the raw data of every pixel.

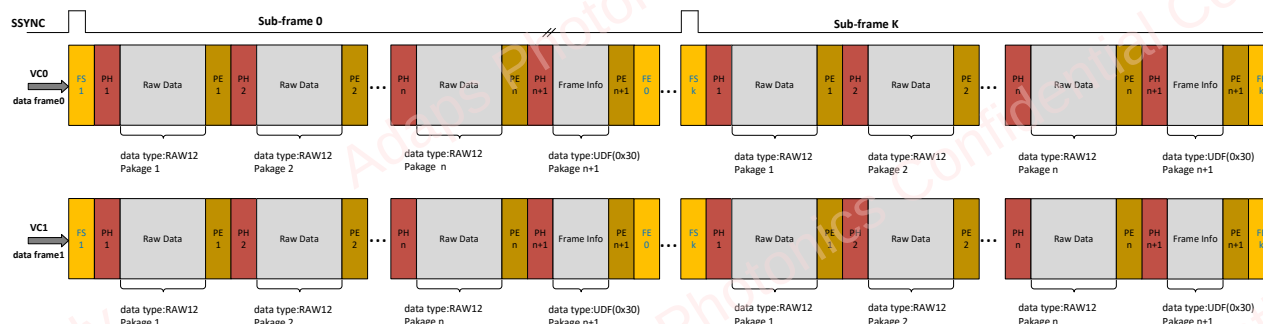


Figure 10-5 data frame format in histogram mode

Note:

- 1) The sub-frame number K is equal to $V_ROLL_NUM@1D$ scan or $V_ROLL_NUM * H_ROLL_NUM@2D$ scan.
- 2) The packet number n depends on the ROI configuration.
- 3) There are two data types in one data frame (0x2C and 0x30).
- 4) On the MIPI bus, one packet of VC1 is transmitted first, then one packet of VC0 later, and the two VCs packet alternate back and forth.

11 System connection example

As the Rx sensor chip in LIDAR system, there are 2 possible system schemes. The main difference is the HOST. Customer can choose FPGA or SoC as the host in the system. But considering the cost and software development convenience, and the accurate timing control is not required for the sensor chip, the SoC scheme is recommended.

11.1 SoC scheme(Recommended)

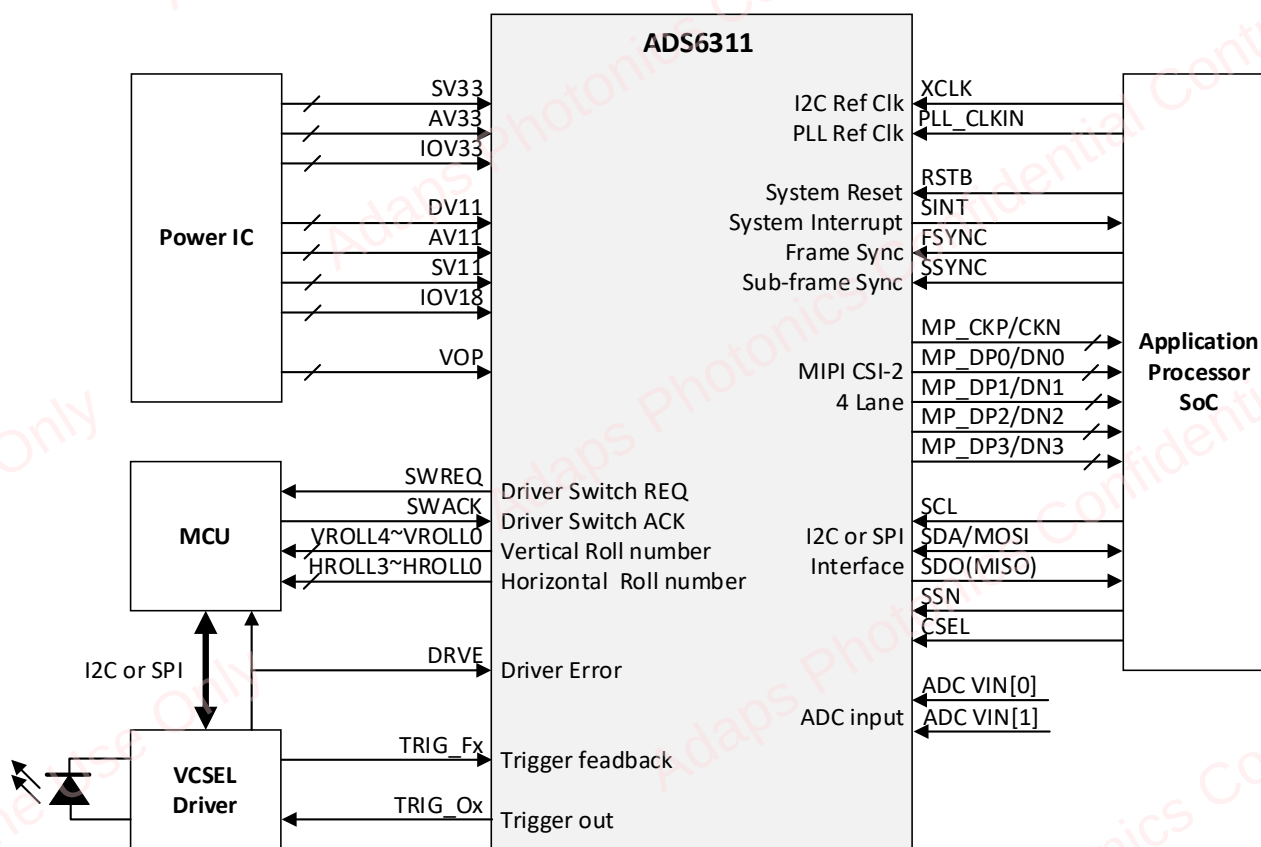


Figure 11-1 SoC scheme block diagram

Application note:

- **MCU:** The main function of MCU is to control the VCSEL driver(s) to switch the different VCSEL lighting zone. It receives monitors the SWREQ, HROLL and VROLL signals from ADS6311, then controls and switches the VCSEL driver configurations. MCU asserts the SWACK signal after VCSEL lighting zone switched. Cortex-M0 series MCUs are satisfied. The CPU frequency should preferably exceed 64MHz. The interface between MCU and VCSEL driver(s) is depend on the driver(s), it's I2C or SPI generally.
- **SoC:** The main function of SoC is to control the sensor and receive the MIPI data then process them. Ideally, the output of SoC should be point cloud data through ethernet interface. The compute power should be no less than 1 TOPs. Four MIPI-CSI 2 data lanes with 1.5Gbps/lane speed are required. The RK3568 chip is recommended.
- The FSYNC and SSYNC signals are used to synchronize with other sensors (e.g. CIS) or HOST. If ADS6311 used in master mode, the FSYNC and SSYNC are output. Otherwise, the HOST should generate the FSYNC and SSYNC signal, meanwhile, the TSYNC timing requirement(Table 9-1 The rolling timing constraints)

should be satisfied.

- The NTC placed near the ADS6311 can be used to measure the sensor chip and PCB temperature. ADC data can be read through I2C/SPI register or MIPI data.
- The TRIG_Fx signal is used for timestamp T0. Which TRIG_Fx (TRIG_F0~TRIG_F3) signals are used depend on customers' Tx scheme.
- Considering the high current referred from Table 7-6 Operating average current characteristics, the Power IC and PCB layout should be carefully evaluated.

11.2 FPGA scheme

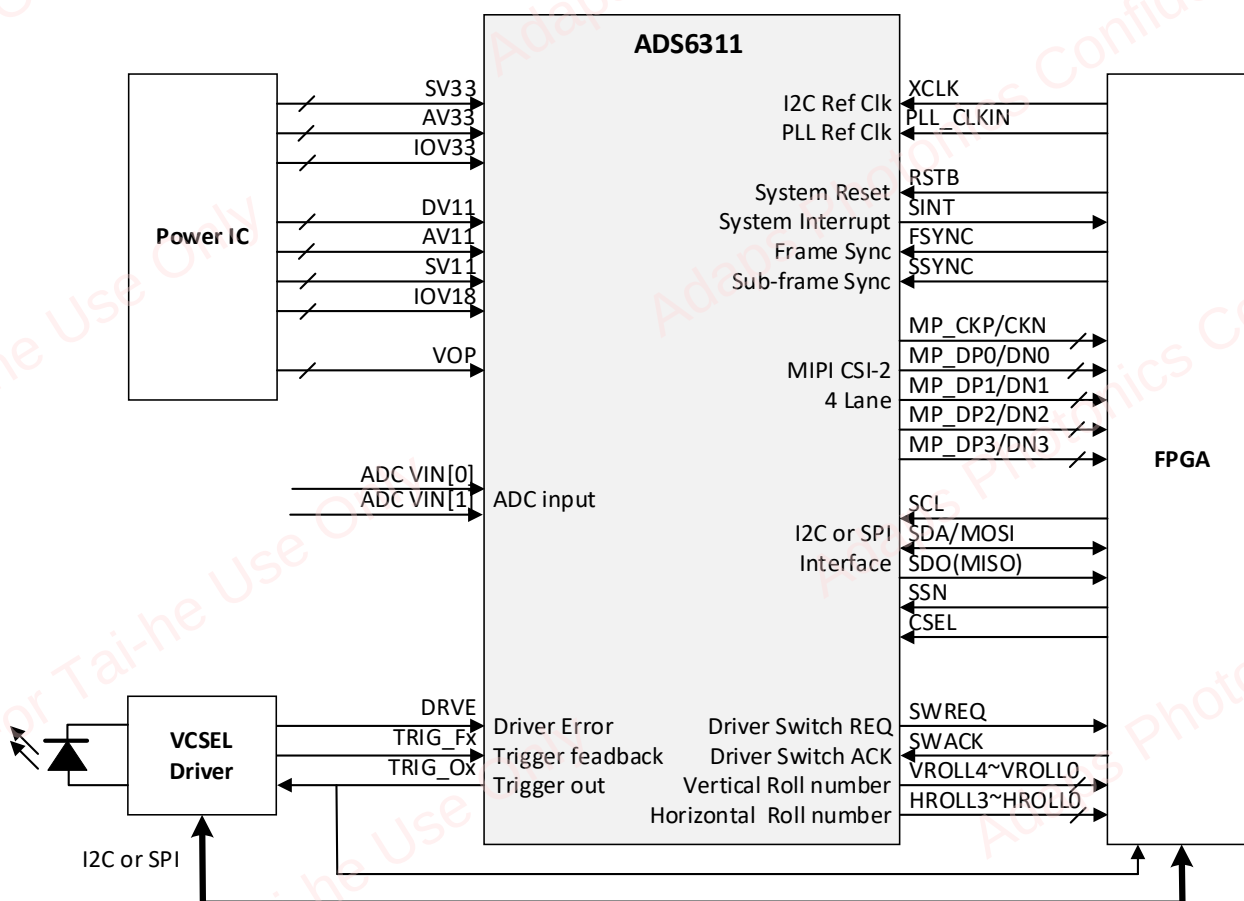


Figure 11-2 FPGA scheme block diagram

Application note:

- The FPGA used as logic control and MIPI data receiving and processing. It combines the function of MCU and SoC in the former scheme. From the perspective of BOM and block diagram, this FPGA scheme is more concise. But considering the MIPI CSI-2 interface and high compute power requirement, this FPGA

will be more expensive.

- The timing requirement and control method is similar with the former one.

12 Ordering Information

Table 12-1 Ordering information

Ordering Code	Package	Automotive/Consumer	uLens	Temperature
ADS6311C06UCD0T	Bare DIE/COB	C	Yes	Tj= -40°C to 125°C
ADS6311A06UFB1T	BGA	A	Yes	Tj= -40°C to 125°C